Group Project

Report due on 2026.11.20, 23:59 Singapore time. Upload to Project under the Assignment section on CANVAS. LLMs may be used for language editing when preparing the report, but the ideas, analysis, and reasoning must originate from the project group.

I. Objective

The objective of this software-based project is to design, implement, and compare different image-processing and machine-learning approaches for the recognition of characters in a microchip image, which is shown in Figure 1.



Fig. 1. Image of a label on a microchip. The image file charact2.bmp is available on CANVAS.

II. Tasks

The following seven tasks are to be completed. **Tasks 1–6** relate to topics covered in Part 1 of the course, whereas **Task 7** relates to topics covered in Part 2.

1. Display the original image on screen. Experiment with contrast enhancement of the image. Comment on the results.
2. Implement and apply a 5×5 averaging filter to the image. Experiment with filters of different sizes. Compare and comment on the results of the respective image smoothing methods.
3. Create a sub-image that includes the line HD44780A00.
4. Convert the sub-image from Task 3 into a binary image.
5. Determine the outline(s) of characters in the image.
6. Segment the image to separate and label the different characters as clearly as possible.
7. A data set contained in the file dataset_2026.zip is provided on CANVAS. Divide this dataset into two portions: (a) a 75% portion that will be used as the training set, and (b) the remaining 25% portion as the test set. Use these two sub-datasets to complete the following three subtasks:
   - *Subtask 7-1*: Design a CNN to classify each character shown in Figure 1.
   - *Subtask 7-2*: Design a classification system, using a non-CNN-based method (or a combination of such methods) selected from those methods that have been covered in Part 2 of this course, to classify each character shown in Figure 1.
   - *Subtask 7-3*: Report the results obtained from *Subtask 7-1* and *Subtask 7-2*, and compare the effectiveness and efficiency of the two approaches (i.e., one uses a CNN, the other does not). Provide your own explanation on any differences in the results between these two approaches.
   - In carrying out Task **7**, also experiment with preprocessing of the data (e.g., padding/resizing the input images) as well as with hyperparameter tuning. Discuss your findings and how sensitive your approach is to these changes in the report. (Note: Do not use Figure 1 as training data for your classifier.)

III. Organization

The class will be divided into groups to work on the project. Each group will be assigned a group number. Each group is required to submit a project report and the corresponding implementation code. The project will be assessed based on these submissions, and all members of the group will receive the same mark.

The implementation code for this project must be developed using either MATLAB[1] or Python. MATLAB's Image Processing Toolbox and other toolboxes related to machine learning may be used to complete the tasks described in Section II.[2] However, students are strongly encouraged to *develop and implement their own code* rather than relying extensively on readily available software packages.

IV. Report Submission

The report should include (i) a title page containing the **names and student numbers of all group members**, (ii) an introduction to the problem, (iii) a description of the algorithms and their flow charts, (vi) screen captures of the intermediate results at each image-processing stag, and (v) a discussion and conclusion describing the rationale behind the chosen method, the investigations conducted, and the lessons learned.

The report should be prepared using 11-point Arial font and single-line spacing, with a maximum length of 50 pages. A typical report should be 30–40 pages long. To facilitate project grading, the report must include a step-by-step guide explaining how to run the implementation code. It is *not the responsibility of the grader* to determine how to configure the software environment required to execute the code.

Put the report and the MATLAB or Python code into a folder and create a zipfile with the name Group_XX.zip, where "XX" stands for your group number (e.g., 01). Upload the file Group_XX.zip to Project under Assignments on CANVAS.

V. Evaluation

Main factors considered in project grading include: (i) the extent to which project work addresses the requirements and produces technically sound and accurate results; (ii) the effectiveness, robustness, and overall standard of the proposed approach and its implementation; (iii) the level of commitment demonstrated, including the scope of work, the depth of investigation, and the initiative shown in overcoming challenges; (iv) the clarity and organization of the written report, including the presentation and discussion of the methodology, results, and conclusions.

[1] https://nusit.nus.edu.sg/services/software_and_os/software/software-for-student/

[2] All external sources, libraries, toolboxes, and third-party code must be properly acknowledged and referenced in the report.